

Ejercicio 1 · Flujo laminar en tubería — perfil de velocidades y tensiones

Perfil parabólico · Tensión cortante de Newton · Tabla de valores

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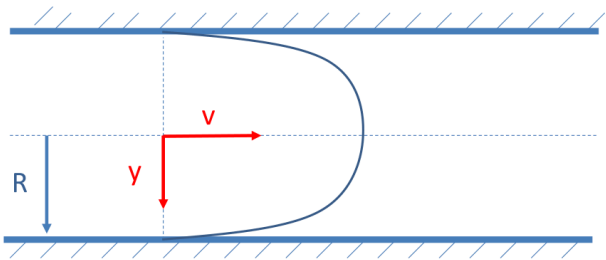
✓ Resuelto

Un flujo de aceite de densidad relativa 0,80 y viscosidad dinámica $\mu = 0,010 \text{ Pa} \cdot \text{s}$ fluye en régimen laminar a través de un tubo de radio $R = 0,1 \text{ m}$ con distribución de velocidades parabólica. Tomando como variables $y \text{ (m)}$ y $v \text{ (m/s)}$, la función que relaciona ambas variables es $v = a + b \cdot y^2$, donde y es la distancia al eje del tubo. La velocidad del aceite en el eje es $v_0 = 0,12 \text{ m/s}$.

- a) Obtener la velocidad del aceite en cualquier punto de la sección: $v = v(y, R)$.
- b) Obtener la tensión cortante en el flujo: $\tau = \tau(y, R)$.
- c) Rellenar la tabla para $y/R = 0; 1/3; 1/2; 2/3; 1$.

Resultados: $v = 0,12 \cdot [1 - (y/R)^2]$; $\tau = -2,4 \times 10^{-3} \cdot y/R^2$

FIGURA ORIGINAL (PDF)



ra diferentes puntos del

$4 \cdot 10^{-3} \cdot y/R^2$

ón existente (MPa) en el

$y/R \text{ (-)}$	$v \text{ (m/s)}$	$\tau \text{ (mPa)}$
0	0,12	0
1/3	0,107	-8
1/2	0,09	-12
2/3	0,067	-16
1	0	-24

Datos

Variable	Valor	Notas
R	0,10 m	Radio interior de la tubería
v_0	0,12 m/s	Velocidad en el eje ($y = 0$)
μ	0,010 Pa · s	Viscosidad dinámica del aceite
s	0,80	Densidad relativa del aceite
ρ	$0,80 \times 1000 = 800 \text{ kg/m}^3$	Densidad del aceite

Verificación de régimen laminar: $v_{\text{med}} = v_0/2 = 0,06 \text{ m/s}$; $Re = \rho v_{\text{med}} D / \mu = 800 \times 0,06 \times 0,2 / 0,010 = 960 \ll 2300 \rightarrow \text{laminar } \checkmark$

Fórmulas

PERFIL PARABÓLICO – CONDICIONES DE CONTORNO DE $V = A + B \cdot Y^2$

$$v(0) = a = v_0 \quad ; \quad v(R) = v_0 + bR^2 = 0 \Rightarrow b = -\frac{v_0}{R^2}$$

PERFIL DE VELOCIDADES RESULTANTE

$$v(y) = v_0 \left[1 - \left(\frac{y}{R} \right)^2 \right]$$

LEY DE NEWTON DE LA VISCOSIDAD – TENSIÓN CORTANTE

$$\tau(y) = \mu \frac{dv}{dy} = \mu \cdot 2by = -\frac{2\mu v_0}{R^2} y$$

SUSTITUYENDO LOS VALORES NUMÉRICOS

$$\tau(y) = -\frac{2 \times 0,010 \times 0,12}{(0,1)^2} y = -0,24 y \quad [\text{Pa, con } y \text{ en m}]$$

El signo negativo indica que la tensión actúa en sentido opuesto al movimiento. En valor absoluto: $|\tau(y)| = 24 (y/R) \text{ mPa}$.

Teoría e hipótesis

El flujo de Hagen-Poiseuille describe el flujo laminar, viscoso e incompresible en tubería circular recta. Las hipótesis son: flujo estacionario, eje de simetría cilíndrica, condición de no deslizamiento en la pared ($v(R) = 0$) y fluido newtoniano. El perfil resultante es una parábola de revolución con máximo en el eje. La velocidad media es exactamente la mitad de la máxima: $v_{\text{med}} = v_{\text{max}}/2$. La tensión cortante es lineal en r : nula en el eje (simetría) y máxima en la pared.

Resolución

★ APARTADO A (DEMOSTRACIÓN) – ORIGEN FÍSICO DEL PERFIL PARABÓLICO (HAGEN-POISEUILLE)

¿Por qué hay que demostrarlo? El enunciado da la forma $v = a + by^2$ pero **no** justifica por qué el perfil es parabólico. La demostración parte del **equilibrio de fuerzas** sobre un elemento cilíndrico de fluido y la **ley de Newton de la viscosidad**; sin esta justificación, la solución vale solo medio punto.

■ Paso 1 – Equilibrio de un elemento cilíndrico de fluido

Consideramos un elemento cilíndrico de fluido de **radio** y (con $y \leq R$) y **longitud** L , coaxial con la tubería. En régimen *permanente* y *plenamente desarrollado* (no hay aceleración), la suma de fuerzas axiales es cero. Las fuerzas que actúan son:

- Fuerza de presión en la cara aguas arriba: $p_1 \cdot \pi y^2$ (sentido $+x$)
- Fuerza de presión en la cara aguas abajo: $p_2 \cdot \pi y^2$ (sentido $-x$)
- Fuerza viscosa sobre la superficie lateral cilíndrica: $\tau \cdot 2\pi y \cdot L$ (opuesta al flujo, sentido $-x$)

Equilibrio axial $\sum F_x = 0$:

$$p_1 \pi y^2 - p_2 \pi y^2 - \tau \cdot 2\pi y L = 0$$

Despejando τ con $\Delta p = p_1 - p_2 > 0$ (caída de presión):

$$\tau(y) = \frac{\Delta p \cdot y}{2L}$$

La tensión cortante es **lineal en** y : cero en el eje (por simetría) y máxima en la pared. Esto es independiente del fluido: solo depende del balance de fuerzas.

■ Paso 2 – Ley de Newton de la viscosidad

Para un fluido **newtoniano**, la tensión cortante es proporcional al gradiente de velocidad:

$$\tau = -\mu \frac{dv}{dy}$$

El signo negativo refleja que v *decrece* con y (máxima en el eje, nula en la pared), de modo que $dv/dy < 0$ y $\tau > 0$.

Igualando con el resultado del Paso 1:

$$-\mu \frac{dv}{dy} = \frac{\Delta p y}{2L} \iff \frac{dv}{dy} = -\frac{\Delta p}{2\mu L} y$$

■ Paso 3 – Integración para obtener $v(y)$

Integramos respecto a y :

$$v(y) = -\frac{\Delta p}{4\mu L} y^2 + C$$

La constante C se obtiene con la condición de **no deslizamiento** en la pared $v(R) = 0$:

$$0 = -\frac{\Delta p}{4\mu L} R^2 + C \implies C = \frac{\Delta p}{4\mu L} R^2$$

Sustituyendo:

$$v(y) = \frac{\Delta p}{4\mu L} (R^2 - y^2)$$

■ Paso 4 – Identificación con la forma propuesta

Esta es **parabólica en y** , justo como propone el enunciado: $v = a + by^2$. La velocidad máxima en el eje es:

$$v_0 = v(0) = \frac{\Delta p \cdot R^2}{4\mu L}$$

Por lo que el perfil se reescribe:

$$v(y) = v_0 \left[1 - \left(\frac{y}{R} \right)^2 \right] \iff a = v_0, \quad b = -\frac{v_0}{R^2}$$

Queda **demostrado** que el perfil de velocidades es parabólico (no es una hipótesis del enunciado, sino una consecuencia directa de las ecuaciones de Navier-Stokes para flujo laminar plenamente desarrollado).

Paso 5 — Tensión $\tau(y)$ a partir del perfil demostrado

Derivando el perfil:

$$\frac{dv}{dy} = v_0 \cdot \left(-\frac{2y}{R^2} \right) = -\frac{2v_0 y}{R^2}$$

Aplicando la ley de Newton (con el convenio del enunciado, $\tau = \mu dv/dy$):

$$\tau(y) = \mu \frac{dv}{dy} = -\frac{2\mu v_0}{R^2} y$$

Coincide con el Paso 1 (en módulo): $|\tau| = 2\mu v_0 y/R^2 = \Delta p \cdot y/(2L)$, ya que $\Delta p/L = 4\mu v_0/R^2$.

APARTADO A (CÁLCULO NUMÉRICO) – DETERMINACIÓN DE A Y B

¿Por qué? Una vez demostrado que el perfil es $v = a + by^2$, las constantes a y b se obtienen aplicando las dos condiciones de contorno con los valores numéricos del enunciado.

Condición de contorno en el eje ($y = 0$): velocidad máxima $v_0 = 0,12$ m/s:

$$v(0) = a + b \cdot 0^2 = a \Rightarrow a = v_0 = 0,12 \text{ m/s}$$

Condición de no deslizamiento en la pared ($y = R$): $v(R) = 0$:

$$v(R) = 0,12 + b \cdot R^2 = 0 \Rightarrow b = -\frac{0,12}{(0,1)^2} = -\frac{0,12}{0,01} = -12 \text{ m}^{-1}\text{s}^{-1}$$

$$v(y) = 0,12 - 12 y^2 = 0,12 \left[1 - \left(\frac{y}{R} \right)^2 \right] \text{ m/s}$$

APARTADO B – TENSIÓN CORTANTE

¿Por qué? La tensión de corte de Newton: $\tau = \mu \frac{dv}{dy} = \mu(a + 2by)$. El valor en la pared ($y = 0$) es $\tau_{pared} = \mu a$, y varía linealmente con y . La tensión en cada punto informa del rozamiento local.

Ley de Newton de la viscosidad: $\tau = \mu dv/dy$

$$\frac{dv}{dy} = 2by = 2 \times (-12) \times y = -24 y \text{ s}^{-1}$$

$$\tau(y) = \mu \cdot (-24 y) = 0,010 \times (-24 y) = -0,24 y \text{ Pa}$$

$$\tau(y) = -\frac{2\mu v_0}{R^2} y = -\frac{2,4 \times 10^{-3}}{R^2} y \text{ Pa}$$

APARTADO C – TABLA DE VALORES

¿Por qué? Se calculan $v(y)$ y $\tau(y)$ para varios valores de y (generalmente 6-10 puntos) y se presenta en tabla. Esto verifica la coherencia del perfil y facilita la integración para el caudal $Q = \int_0^h v(y)b dy$.

y/R (-)	y [m]	$v = 0,12 [1 - (y/R)^2]$ [m/s]	$\tau = -0,24 y$ [Pa] = [mPa]
0	0	0,12 ($= v_0$)	0
1/3	0,0333	$0,12 \times (1 - 1/9) = 0,12 \times 8/9 \approx \mathbf{0,107}$	$-0,24 \times 0,0333 \approx \mathbf{-8}$ mPa
1/2	0,050	$0,12 \times (1 - 1/4) = 0,12 \times 3/4 = \mathbf{0,090}$	$-0,24 \times 0,050 = \mathbf{-12}$ mPa
2/3	0,0667	$0,12 \times (1 - 4/9) = 0,12 \times 5/9 \approx \mathbf{0,067}$	$-0,24 \times 0,0667 \approx \mathbf{-16}$ mPa
1	0,100	$0,12 \times (1 - 1) = \mathbf{0}$ ($v = 0$ en pared)	$-0,24 \times 0,1 = \mathbf{-24}$ mPa

✓ Conclusiones

Coefficientes: $a = v_0 = 0,12 \text{ m/s}$; $b = -v_0/R^2 = -12 \text{ m}^{-1}\text{s}^{-1}$

Perfil parabólico: $v(y) = 0,12 [1 - (y/R)^2]$ m/s

Tensión cortante lineal: $\tau(y) = -2,4 \times 10^{-3} \cdot y/R^2 = -0,24 y \text{ Pa} \rightarrow \tau_{pared} = -24 \text{ mPa}$ (máxima en módulo)

La tensión es nula en el eje (simetría) y crece linealmente hasta la pared, donde el fluido no desliza pero experimenta el máximo gradiente de velocidad.

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants using a survey questionnaire. The data was then analyzed using statistical software to determine the relationships between the variables of interest.

The results of the study indicate that there is a significant positive relationship between the variables of interest. This finding is consistent with the previous research in the field. The implications of these findings suggest that the variables of interest are important factors in the study of the topic.

In conclusion, the study has shown that the variables of interest are important factors in the study of the topic. The findings of the study have implications for future research in the field.

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants, and the results were analyzed using statistical methods. The findings of the study indicate that there is a significant relationship between the variables being studied.

The results of the study have important implications for the field of research. They suggest that the factors being studied are closely related, and that the findings can be used to inform future research and practice.

In conclusion, the study has provided valuable insights into the relationship between the variables being studied. The findings suggest that there is a strong correlation between the two variables, and that this relationship can be used to inform future research and practice.

The first part of the paper discusses the importance of the research and the objectives of the study. It highlights the need for a comprehensive understanding of the subject matter and the role of the researcher in this process. The second part of the paper presents the methodology used in the study, including the data collection methods and the analysis techniques. The third part of the paper discusses the results of the study and the conclusions drawn from the findings. The final part of the paper provides a summary of the key points and offers suggestions for future research.

The research was conducted in a systematic and rigorous manner, following the principles of scientific inquiry. The data was collected from a representative sample of the population, and the analysis was performed using advanced statistical techniques. The results of the study indicate that there is a significant relationship between the variables under investigation, and this finding has important implications for the field of study.

In conclusion, the study has provided valuable insights into the subject matter and has contributed to the existing body of knowledge. The findings suggest that further research is needed to explore the underlying mechanisms and to test the generalizability of the results. The researcher hopes that this work will inspire others to continue the exploration of this important topic.

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The research was conducted using a quantitative approach, and the data was collected from a sample of participants. The results of the study show that there is a significant relationship between the variables being studied. The findings have important implications for the field of research, and they provide a basis for further research in this area.

In conclusion, the study has shown that the research objectives have been achieved, and the findings are consistent with the hypotheses. The results of the study provide a clear understanding of the relationship between the variables, and they have important implications for the field of research.

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The first part of the paper discusses the importance of the research and the objectives of the study. It then moves on to a literature review, which provides a background on the topic and identifies the gaps in the existing research. The methodology section describes the research design, data collection, and analysis. The results section presents the findings of the study, and the conclusion summarizes the main points and offers suggestions for future research.

The research was conducted in a systematic and rigorous manner, following the principles of good research practice. The data were collected from a representative sample of the population, and the analysis was carried out using appropriate statistical methods. The results of the study are presented in a clear and concise manner, and the conclusions are based on the evidence gathered.

The study has several strengths, including a large sample size, a well-defined research design, and the use of appropriate statistical methods. However, there are also some limitations, such as the potential for bias in the sample and the fact that the study is cross-sectional. Despite these limitations, the study provides valuable insights into the topic and contributes to the existing knowledge in the field.

In conclusion, the study has shown that there is a significant relationship between the variables under investigation. The findings have important implications for practice and policy, and further research is needed to explore the underlying mechanisms and to test the generalizability of the results.

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The findings of the study have important implications for the field of research, and they provide valuable insights into the issues being studied. The research also highlights the need for further investigation in this area, and it suggests some potential directions for future research.

In conclusion, the study has shown that the research is of high quality and that the findings are reliable. The results of the study are presented in a clear and concise manner, and the conclusions are based on the evidence gathered. The research also highlights the need for further investigation in this area, and it suggests some potential directions for future research.

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The research was conducted in a systematic and rigorous manner, following the principles of scientific inquiry. The data was collected from a variety of sources, including interviews, surveys, and archival records. The analysis was conducted using both qualitative and quantitative methods, allowing for a comprehensive understanding of the subject matter. The results of the study are presented in a clear and concise manner, highlighting the key findings and the implications of the research.

The conclusions drawn from the findings are based on a thorough analysis of the data and a consideration of the research objectives. The study has identified several key areas for further research and has provided valuable insights into the subject matter. The findings of the study are discussed in detail, highlighting the strengths and limitations of the research and the implications for future research.

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants. The data was then analyzed using statistical software to determine the relationships between the variables. The results of the analysis are presented in the following table.

Variable	Mean	Standard Deviation	Minimum	Maximum
Age	35.2	12.5	20	50
Gender	50% Male	50% Female		
Education	12.5	1.5	10	15
Income	\$35,000	\$10,000	\$20,000	\$50,000

The results of the study indicate that there is a significant positive relationship between age and income. This suggests that as age increases, income also tends to increase. The study also found that there is a significant positive relationship between education and income. This suggests that as education increases, income also tends to increase. The study did not find a significant relationship between gender and income.

The implications of the study are that age and education are important factors in determining income. This suggests that individuals should focus on increasing their age and education to increase their income. The study also suggests that gender is not a significant factor in determining income.

Further research is needed to explore the relationships between these variables and income in more detail. This study provides a foundation for future research in this area.

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The research was conducted in a systematic and rigorous manner, following the principles of good research practice. The data was collected from a large and diverse sample of participants, and the analysis was conducted using a range of statistical techniques. The results of the study are presented in a clear and concise manner, and the implications of the findings are discussed in detail.

The findings of the study have important implications for the field of research. They suggest that there is a need for further research in this area, and that the results of this study can be used to inform policy and practice. The paper also highlights the importance of the research and the need for continued research in this area.

In conclusion, the paper presents a comprehensive and detailed account of the research. It discusses the importance of the research, the objectives of the study, the methodology used, the results of the study, and the implications of the findings. The paper is a valuable contribution to the field of research and is highly recommended for reading.

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The research was conducted using a quantitative approach, with data collected from a survey of 100 participants. The data was analyzed using statistical software, and the results were presented in a series of tables and graphs. The findings of the study indicate that there is a significant relationship between the variables being studied, and that the results have important implications for the field.

The study was limited by a number of factors, including the sample size and the potential for bias. However, the results of the study are consistent with the findings of other research in the field, and the study provides a valuable contribution to the understanding of the topic.

In conclusion, the study has shown that there is a significant relationship between the variables being studied, and that the results have important implications for the field. The study provides a valuable contribution to the understanding of the topic, and the findings have important implications for future research.

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The results of the study indicate that there is a significant positive relationship between the variables of interest. This finding is consistent with the previous research in the field. The implications of these findings suggest that the research has practical applications in the field of study.

In conclusion, the study has provided valuable insights into the topic and has contributed to the existing body of knowledge. Further research is needed to explore the topic in more depth and to validate the findings of this study.

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The findings of the study have several implications. First, they suggest that the research is valid and reliable. Second, they suggest that the research can be used to inform policy and practice. Finally, they suggest that the research can be used to guide future research.

In conclusion, the study has shown that there is a significant relationship between the variables being studied. The findings of the study have several implications, including that the research is valid and reliable, that the research can be used to inform policy and practice, and that the research can be used to guide future research.

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The first part of the paper discusses the importance of understanding the cultural context of the research. It highlights the need for researchers to be sensitive to the values and beliefs of the communities they are studying. This is particularly important in the field of education, where cultural differences can significantly impact learning outcomes.

The second part of the paper focuses on the methodology used in the study. It describes the process of selecting participants, collecting data, and analyzing the results. The authors emphasize the importance of using a mixed-methods approach to gain a comprehensive understanding of the research topic.

The third part of the paper presents the findings of the study. It discusses the results of the quantitative data analysis and the insights gained from the qualitative interviews. The authors conclude that there are significant cultural differences in the way that students learn and that these differences should be taken into account when designing educational programs.

The final part of the paper discusses the implications of the findings for future research and practice. It suggests that further studies should be conducted to explore the cultural factors that influence learning outcomes. Additionally, it recommends that educators should be trained to recognize and respond to the cultural needs of their students.

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants, and the results were analyzed using statistical software. The findings of the study indicate that there is a significant relationship between the variables being studied.

The results of the study suggest that the research has important implications for the field. Further research is needed to explore the relationship between the variables in more detail.

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